

Santiago Coelho

Curriculum Vitae

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Education

- Ph.D. in Biomedical Engineering** 2015–2019
University of Leeds (2018–2019) and University of Sheffield (2015–2018), UK. Advisor: Prof. Alejandro Frangi.
- Bachelor in Electrical Engineering** 2010–2015
Five-year program, Universidad Nacional de La Plata (UNLP), La Plata, Argentina. Engineering Faculty Salutatorian.

Research and Teaching Experience

- Assistant Professor (Research)** Aug. 2026–present
Center for Biomedical Imaging (CBI), Department of Radiology, School of Medicine, New York University.
- Postdoctoral Research Fellow** Jan. 2020–July 2026
MRI Biophysics Group (PIs: Dmitry Novikov and Els Fieremans), Center for Biomedical Imaging (CBI), Department of Radiology, School of Medicine, New York University.
- Teaching Instructor** 2021–present
Practical Magnetic Resonance Imaging II (Ph.D. course), Department of Radiology, School of Medicine, New York University.
- Graduate Teaching Assistant** 2017–2018
Mathematics of Imaging Sciences (M.Sc. course), Department of Electronic and Electrical Engineering, University of Sheffield.
- Teaching Assistant** 2012–2015
Numerical Analysis, Faculty of Engineering, Universidad Nacional de La Plata (FiUNLP).
- Teaching Assistant** 2014
Control and Servomechanisms, FiUNLP.
- Teaching Assistant** 2013
Electromagnetism, FiUNLP.
- Teaching Assistant** 2011
Calculus II, FiUNLP.

Honors and Awards

- **Junior Fellow of the ISMRM** (2025).
- **NIH K99/R00 Award** (K99EB036080-01), NIBIB, NIH; approximately \$1M for 2024–2029.
- Magna Cum Laude Award, ISMRM Annual Meeting, work #0641 (2024).
- Summa Cum Laude Award and AMPC Top 100 selection, ISMRM Annual Meeting, work #0069 (2023).
- Best Presentation Award, ISMRM Diffusion Study Group Workshop (2022).
- Magna Cum Laude Award, ISMRM Annual Meeting, work #0514 (2022).
- Best Presentation Award, ISMRM Diffusion Study Group, ISMRM Annual Meeting (2022).
- Best Presentation Award, ISMRM White Matter Study Group, ISMRM Annual Meeting (2022).
- Summa Cum Laude Award, ISMRM Annual Meeting, work #0397 (2021).
- Best Presentation Award, ISMRM Diffusion Day Trainee Workshop (2021).
- Award to the Best Graduate of Electrical Engineering, UNLP, La Plata, Argentina (2015).
- Annual Award from the Professional Council of Mechanical and Electrical Engineering for the highest GPA (2015).
- J. V. Gonzalez Award, La Plata City Council, for achieving one of the highest GPAs in FiUNLP (2015).
- Best Graduate Award, FiUNLP, for achieving the highest GPA in an engineering specialty (2015).
- Luis A. Huerco Award, Buenos Aires College of Engineers, for the highest GPA in a UNLP engineering specialty (2015).

Leadership

- **Elected Trainee Representative**, ISMRM Diffusion Study Group (approximately 1,000 members), 2023–2025.
- **Chair**, Tissue Microstructure Imaging Gordon Research Seminar (GRS), 2023.
Held in conjunction with the namesake Gordon Research Conference, the seminar provided a forum for doctoral and postdoctoral researchers to present work, discuss emerging methods and pre-publication data, and build collaborations. It covered multimodal approaches to tissue imaging at the cellular level, including microscopy, ultrasound, and MRI.
- **Co-Chair**, ISMRM Diffusion Trainee Day, 2024.
This 24-hour virtual workshop spanned three time zones and included educational lectures, scientific and poster presentations, and networking activities for early-career researchers.
- **Co-supervisor**, M.Sc. project, “Multiple approaches for Diffusion Tensor Imaging estimation” (2017). Main advisor: Prof. Alejandro Frangi.

Service and Editorial Activities

- **Organizing Committee**, ISMRM Diffusion Study Group Workshop, “40 Years of Diffusion: Past, Present & Future Perspectives”, Kyoto, February 2025. Co-Chair of the Educational and Historical Perspective Programs.
- **Organizing Committee**, ISMRM Member-Initiated Session, “Pushing the Boundaries of Diffusion MRI Across Field Strengths & Gradient Performance” (MIS-10), Singapore, 2024.
- **Guest Editor**, virtual special issue of *Magnetic Resonance in Medicine* on Diffusion MR (2023).
- **Co-Managing Guest Editor**, special issue of *Imaging Neuroscience* on Diffusion MRI (2025–2026).
- **Reviewer** for *NeuroImage*, *Magnetic Resonance in Medicine*, *Imaging Neuroscience*, *Communications Biology*, *Human Brain Mapping*, *Medical Image Analysis*, *NMR in Biomedicine*, *Scientific Reports*, and *American Journal of Neuroradiology*.

Patents

1. **Santiago Coelho**, Els Fieremans, and Dmitry S. Novikov. “System, method and computer-accessible medium for determining rotational invariants of cumulant expansion from one or more acquisitions which can be minimal.” U.S. Provisional Application 63/333,856; converted to a full PCT application April 25, 2023.
2. **Santiago Coelho**, Els Fieremans, and Dmitry S. Novikov. “System, method and computer-accessible medium for diffusion MRI without shells.” Provisional Patent Application 63/412,181; converted to a full PCT application September 30, 2023.

Peer-Reviewed Publications

1. **Santiago Coelho**, Jenny Chen, Filip Szczepankiewicz, Els Fieremans, and Dmitry S. Novikov. “Geometry of the cumulant series in diffusion MRI.” *Nature Communications*, 17:4220, 2026.
2. **Santiago Coelho**, . . . , Thomas K. F. Foo, Els Fieremans, and Dmitry S. Novikov. “What if each voxel were measured with a different diffusion protocol?” *Magnetic Resonance in Medicine*, 2025.
3. Sohae Chung, . . . , **Santiago Coelho**, . . . , and Yvonne W. Lui. “Linking Symptom Phenotypes to Patterns of White Matter Injury in Mild Traumatic Brain Injury: A Latent Class Analysis.” *American Journal of Neuroradiology*, 2025.
4. Kwok-Shing Chan, Yixin Ma, Hansol Lee, José P. Marques, Jonas Olesen, **Santiago Coelho**, et al. “In vivo human neurite exchange time imaging at 500 mT/m diffusion gradients.” *Imaging Neuroscience*, 3:1–26, 2025.
5. Benjamin Ades-Aron, **Santiago Coelho**, et al. “Denoising Improves Cross-Scanner and Cross-Protocol Test–Retest Reproducibility of Diffusion Tensor and Kurtosis Imaging.” *Human Brain Mapping*, 46:e70142, 2025.
6. **Santiago Coelho**, Ying Liao, et al. “Assessment of precision and accuracy of brain white matter microstructure using combined diffusion MRI and relaxometry.” *Human Brain Mapping*, 2024.
7. Ricardo Coronado-Leija, Ali Abdollahzadeh, Hong-Hsi Lee, **Santiago Coelho**, et al. “Volume electron microscopy in injured rat brain validates white matter microstructure metrics from diffusion MRI.” *Imaging Neuroscience*, 2024.
8. Ying Liao, **Santiago Coelho**, Jenny Chen, et al. “Mapping tissue microstructure of brain white matter in vivo in health and disease using diffusion MRI.” *Imaging Neuroscience*, 2024.
9. Sohae Chung, Tamar Bacon, Joseph F. Rath, Alaleh Alivar, **Santiago Coelho**, et al. “Callosal Interhemispheric Communication in Mild Traumatic Brain Injury: A Mediation Analysis on WM Microstructure Effects.” *American Journal of Neuroradiology*, 2024.
10. **Santiago Coelho**, Steven H. Baete, Gregory Lemberskiy, et al. “Reproducibility of the standard model of diffusion in white matter on clinical MRI systems.” *NeuroImage*, 2022.
11. L. A. Wilson, L. Heraty, B. A. Ashford, **Santiago Coelho**, A. F. Frangi, J. M. Pozo, P. G. Ince, and J. R. Highley. “Tissue

microarray (TMA) use in post mortem neuropathology.” *Journal of Neuroscience Methods*, 2021.

12. Meghdoot Mozumder, Jose M. Pozo, **Santiago Coelho**, and Alejandro F. Frangi. “Population-based Bayesian regularization for microstructural diffusion MRI with NODDIDA.” *Magnetic Resonance in Medicine*, 2019.
13. **Santiago Coelho**, Jose M. Pozo, et al. “Histological data of axons, astrocytes, and myelin in deep subcortical white matter populations.” *Data in Brief*, 2019.
14. **Santiago Coelho**, Jose M. Pozo, et al. “Resolving degeneracy in diffusion MRI biophysical model parameter estimation using double diffusion encoding.” *Magnetic Resonance in Medicine*, 2019.
15. Rachel Waller, Lynne Baxter, Daniel J. Fillingham, **Santiago Coelho**, et al. “Iba-1-/CD68+ microglia are a prominent feature of age-associated deep subcortical white matter lesions.” *PLoS ONE*, 2019.
16. Meghdoot Mozumder, Jose M. Pozo, **Santiago Coelho**, et al. “Quantitative histomorphometry of capillary microstructure in deep white matter.” *NeuroImage: Clinical*, 2019.
17. **Santiago Coelho**, Jose M. Pozo, et al. “Local volume fraction distributions of axons, astrocytes, and myelin in deep subcortical white matter.” *NeuroImage*, 2018.

Book Chapters

1. **Santiago Coelho**, Yogesh Rathi, and Marco Reisert. “Model Estimation,” in *Probing Brain Microstructure with MRI*, Springer Nature’s Neuroscience Methods series. Edited by Sune Jespersen and Noam Shemesh; to be published in 2026. *In preparation*.
2. Els Fieremans, Kouhei Kamiya, and **Santiago Coelho**. “Standard Diffusion MRI,” in *Diffusion MRI User Guide: A Window to Biological Tissue*, Springer Nature’s Diagnostic Imaging Series. Edited by Mami Iima, Susie Huang, and Denis Le Bihan; to be published in 2026. *In preparation*.

Preprints and Publications Under Review

1. Valentin Stepanov, **Santiago Coelho**, Benjamin Ades-Aron, Jenny Chen, Timothy Shepherd, Ilya Kister, Dmitry Novikov, and Els Fieremans. “Advanced quantitative diffusion MRI markers unveil changes in the normal-appearing white matter reflecting disability in multiple sclerosis.” MedArxiv, 2025. *Under review*.
2. **Santiago Coelho**, Els Fieremans, and Dmitry S. Novikov. “Zero-Shell Imaging (ZSI): microstructure-informed undersampling of diffusion MRI.” *In preparation; planned for submission in 2026*.
3. **Santiago Coelho**, Els Fieremans, and Dmitry S. Novikov. “How do we know we measure tissue parameter, not the prior?” *In preparation; planned for submission in 2026*.

Presentations at Scientific Conferences

Peer-Reviewed Conference Abstracts

Delivered 40 presentations at ISMRM meetings (2017–2025), including the following first- or senior-author oral presentations and power pitches:

1. **Santiago Coelho**, Els Fieremans, and Dmitry S. Novikov. “When fiber bundles are not axially symmetric: 4-dimensional fiber distributions in the human brain” Annual Meeting of the ISMRM, 2026.
2. Valentina Mazzoli, Smita Rao, . . . , Dmitry S. Novikov, Els Fieremans, and **Santiago Coelho**. “Microstructure and muscle strength in older adults: a time-dependent diffusion study.” Annual Meeting of the ISMRM, 2025.
3. **Santiago Coelho**, Filip Szczepankiewicz, Els Fieremans, and Dmitry S. Novikov. “Cumulant Tensors From The Addition Of Angular Momenta: All Diffusion Invariants In One Abstract.” Annual Meeting of the ISMRM, 2024.
4. **Santiago Coelho**, Valentin Stepanov, et al. “Sticks Or No Sticks? White Matter Microstructure In Multiple Sclerosis From High-B Scaling.” Annual Meeting of the ISMRM, 2024.
5. **Santiago Coelho**, Gregory Lemberskiy, et al. “What If Every Voxel Was Measured With A Different Diffusion Protocol?” Annual Meeting of the ISMRM, 2023.
6. **Santiago Coelho**, Els Fieremans, and Dmitry S. Novikov. “Zero-Shell Diffusion MRI: Focus On Microstructure By Decoupling Fiber Orientations.” Annual Meeting of the ISMRM, 2023.
7. **Santiago Coelho**, Els Fieremans, and Dmitry S. Novikov. “Rotational Invariants Of The Cumulant Expansion (RICE).” Annual Meeting of the ISMRM, 2022.
8. **Santiago Coelho**, Filip Szczepankiewicz, Els Fieremans, and Dmitry S. Novikov. “Fast Acquisitions For Mean Kurtosis And Microscopic Fractional Anisotropy.” Annual Meeting of the ISMRM, 2022.
9. **Santiago Coelho**, Filip Szczepankiewicz, et al. “Reproducibility Of White Matter Microstructure Mapping With Diffusion-

Relaxometry.” Annual Meeting of the ISMRM, 2022.

10. **Santiago Coelho**, Els Fieremans, and Dmitry S. Novikov. “How do we know we measure tissue parameters, not the prior?” Annual Meeting of the ISMRM, 2021.
11. **Santiago Coelho**, et al. “Optimal experimental design in multidimensional diffusion MRI for parameter estimation of biophysical tissue models.” Annual Meeting of the ISMRM, 2020.

Peer-Reviewed Conference Articles

1. **Santiago Coelho**, Jose M. Pozo, Sune N. Jespersen, and Alejandro F. Frangi. “Optimal experimental design for biophysical modelling in multidimensional diffusion MRI.” *MICCAI Proceedings, Lecture Notes in Computer Science*, 2019.

Invited Talks

- **i2i NYU Workshop** — *Invited talk*, New York, October 2026. *Forthcoming*.
- **White Matter Study Group Workshop** — *Educational lecture*, Marseille, France, October 2026. *Forthcoming*.
- **BrainMap Seminars**, Martinos Center, Massachusetts General Hospital and Harvard Medical School — Boston, October 2025.
- **Department of Radiology, Weill Cornell Medicine** — New York, October 2025.
- **What Can You Be With a PhD? Symposium** — *Invited speaker*, New York, October 2025.
- **Institute of Imaging Science, Vanderbilt University** — Nashville, September 2025.
- **Department of Radiology, Columbia University** — New York, August 2025.
- **BioMedical Engineering and Imaging Institute, Icahn School of Medicine at Mount Sinai** — New York, April 2025.
- **Diffusion Microclub Scientific Seminars** — Virtual, October 2024.
- **ISMRM Annual Meeting** — *Educational lecture*, Toronto, Canada, June 2023.
- **ISMRM Diffusion Study Group Virtual Workshop** — *Educational lecture*, December 2021.
- **ISMRM Annual Meeting** — *Educational lecture*, virtual, May 2021.
- **Faculty of Engineering, Universidad Nacional de La Plata** — La Plata, Argentina, December 2019.